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Terrestrial analogues to Mars, the Moon and microgravity: Analysis of research sites in Colombia as an emerging country in space activities

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Abstract

As human space exploration advances, it becomes necessary to test new technologies, procedures, and in general how some parts of a mission will be developed, from hardware performance, to logistics and operations, including the analysis of team dynamics, individual psychology and physiology, human factors during a prolonged isolation time as well as limited and delayed communication. Terrestrial analogue missions are designed to have similar geological, environmental and/or biological conditions as in extreme space environments. Current analogue missions, focused on simulating possible scenarios comparable to the surface of Mars or the Moon, require locations that imitate certain aspects of said surfaces for the tests in mention. Mission constraints and operational scenarios depend on the characteristics of the analogue sites. Colombia has a lot to offer in this aspect, given that its geography offers multiple possibilities for analogue mission sites ranging from high-relief to smooth and flat deserts, volcano edifices similar to lunar terrain, hot springs hosting endemic species, caves resembling lunar or martian lava tubes and snow top mountains, among others. Also, existing infrastructure such as diving pools, to perform simulated EVAs under neutral buoyancy conditions, or air force planes capable of performing short parabolic flights are considered as places to simulate analogue environments. For this work, possible suitable locations in Colombia are analysed, characterized and categorised: For each possible location, the type of analogue mission is defined and potential investigation topics are described. This work shows multiple examples for Colombia, where the current improving socio-political situation has opened new opportunities of access to places that were impossible before due to insecurity. The identification of this analogues provides an opportunity to expand research in space exploration matters for the country and the region.

Keywords: Analogue, Colombia, Micro-Tina, Lunar regolith

1. Introduction

Deep rooted in our existence as human beings is the need to fulfill our curiosity and to explore unknown places. During centuries, scientists and astronomers have focused their thoughts on space, laying the first stepstones for its physical exploration with Sputnik 1 in 1957. Since then, multiple robotic missions have been performed not only to Low Earth Orbit but also to other celestial bodies in our solar system. Besides, the presence of human beings on the Moon and a permanent manned settlement in space have brought further human space exploration. Future missions envision a permanent settlement on the Moon and a visit to Mars or asteroids. The design and preparation of such missions will closely be related on the capability of testing hardware and operations and, training astronauts in analogue environments. In these locations, new technologies can be partly validated, procedures can be verified and effects on the body, both physical and mental, can be studied playing an important role in problem solving for spaceflight research.

This work aims to present a first approach of a catalogue of possible sites to perform analogue missions and experiments within Colombia. This country has a varied geography, which provides ample options on sites to test different aspects of a mission, and, the fact of a recent peace treaty makes available sites previously unavailable due to civil war, and allows military equipment to be used as well.

2. Analogue Environments

The characteristics of different analogue environments depend in the simulation goal they want to achieve. Therefore, analogue missions can range from simple laboratory settings to large multinational missions conducted in challenging environments [1], making their categorization challenging. Analogues on Earth can be categorized on different mission aspects. So far, analogue planetary research could be distinguished into two different fields of study; astronaut training and hardware testing [2]. Additionally, analogue missions can be categorized depending on the initiator, be it a space agency or a private organization, as well as the mission duration,

long term and short-term missions. Finally, analogue mission can be categorized into different types of investigation environments as follows:

2.1. Hot deserts

The large uninhabited areas in deserts offer the possibility to build habitats and perform missions on dry and arid grounds. Although vegetation is scarce, a variety of microorganisms and radiotolerant bacteria can be integrated in analogue research goals. Additionally, the geology at deserts can vary from high-relief to smooth and flat deserts, giving a broad variety of surfaces for a field training or hardware test. The goal of analogue missions in hot desert is to take advantage of the on extreme environments while using a site with similar surface characteristics to Mars or the Moon. The red sandstones in some deserts even give a visual similarity to a martian surface. One example is the desert of Utah, where the Mars Desert Research Site (MDRS) has been built in the early 2000s and where small crews visit the site for short periods of time to conduct scientific research.

2.2. Cold deserts

Cold deserts, situated at high latitudes or high altitudes, are usually also places far from any civilization. Cold deserts in Antarctica or the Arctic, as well as snow topped mountains provide appropriate similarities to weathering processes in other planetary bodies, low temperature and reduced humidity provide an environment challenging for different types of hardware to be tested, as well as for survival equipment and training. [3]

2.3. Lakes and Low depth aquatic environments

Lakes and shallow water environments, whether natural or manmade, provide test grounds for several types of research. Systems such as underwater habitats have been tested, which serve to train astronauts for isolation and confinement. Some of these also provide extreme conditions for life, providing similar conditions to those found on other planets, or early stages of formation of planetary bodies. One example for this type of analogue site is the NEEMO Project in which NASA utilizes an aquarius undersea habitat as an analog for long duration flights. [4]

2.4. Vulcanos

Volcanic environments are generally characterized by arid, intricate environments consistent of relatively new rock material, as such, they provide a topography

similar to those found in other planetary bodies surfaces, as well as similar materials as those found on Lunar and martian regolith, consistent mainly on meteorized basaltic rocks. Different types of research is performed in such environments, from entire analogue missions, to hardware testing.

The volcano of Mauna Loa on the island of Hawai is one example for a place where NASA has performed analogue studies.[5]

2.5. Caves

Similar to environments found on the Moon, and Mars, which are suggested as alternatives for habitat set up, reducing radiation risk, as well as providing shelter for the elements. Provide challenges for technology testing, habitat setup, and astronaut training. The ESA CAVES analog site in Sardinia, has been used for the last years to train astronauts. [6]

2.6. Underwater confined facilities

Existing infrastructure such as diving pools provide an adequate environment for low gravity training, as well as EVA procedures for zero gravity operations, such as procedures, hardware handling, and tool usage. It is also an important resource to test aquatic hardware on confinement and controlled conditions, before sending it to extreme environments. The neutral buoyancy facility in Cologne Germany is one example of an underwater confined facilities.

2.7. Parabolic flights

Parabolic flight in planes provides the opportunity to test experiments on different gravity ranges, generally between 0g to 2g, depending on the flight profile. Commercial and military aircrafts, with particular modifications, have been historically used to simulate variable gravity conditions, existing nowadays companies that provide the service as a commercial venture for research and tourism. One example for such a commercial provider is the Zero Gravity Corporation, based in Virginia, USA which offer flights with a Boeing 727-227F. [7]

2.8. High altitude atmosphere

The high-altitude atmosphere provides a proving ground for operations on the surface of planetary bodies with reduced atmospheric density, in particular Mars. Tests of control behaviour and atmospheric sampling have been conducted in the past. One example are high altitude balloons such as the ESA Educational balloons BEXUS.

2.9. *Isolated and confined facility*

Confined facilities can vary widely and have several purposes, from small laboratory equipment to dedicated facilities. The investigation purpose on confined and isolated facilities are principally on individual's physiology and team dynamics. One example for an analog mission performed during a significant amount of time in an analogue facility is Mars 500. [8]

2.10. *Bedrest*

Bedrest tests consist of prolonged periods of time where a person stays in a horizontal position, or a low angle. This condition provides a similarity to low or zero gravity environment, promoting a redistribution of bodily fluids. This kind of study has the advantage that it doesn't require intricate infrastructure, as it can be performed in preexisting medical facilities. An example of a facility where continuous bedrest studies are performed is the ENVIHAB in Cologne, Germany. [9]

The study of analogue environments not also serves the purpose of preparing crewed and uncrewed spaceflight, studying the past aqueous and present arid environments it is possible to gain better understanding on the phenomena that led to those formations, and understand better the conditions on past environments on planetary surfaces.

It is always important to be aware of what element within the environment is the analogue to the space environment, and in that way, to design properly the experiments to be conducted, for an appropriate validation of the results.

3. **Analogue sites in Colombia**

Current analogue sites involve research from multiple disciplines, having scientists, engineers, medical personnel but often also journalists and students being part of a mission. Additionally, international cooperation plays a crucial role in different missions. This interdisciplinary and international context encourage the involvement of developing countries. For Colombia, having people involved in analogue missions means being part of this international cooperation by contributing with research ideas. But besides only having people participating in analogue missions a step further is performing analogue mission at different sites in a near future in Colombia in order to increase the country's capability to contribute towards an international Earth-based field research program

preparing for planetary exploration. Given what the country's geography offers, a variety of environments can be simulated which could be a baseline for a meaningful participation in a global space exploration program. These sites, which are only recently accessible after the current improving socio-political situation, and possible research done on it could lead Colombia towards a path to become an emerging space nation. [10]

4. **Previous analogue studies in Colombia**

Currently, there is a worldwide trend in the increase of analogue studies, in lab, as well as in field. In the case of Colombia, some studies have been made towards it, and while they have not been big scale analogue studies, they underscore the importance and feasibility of this activities in the country. Three of these works are going to be mentioned, the first one corresponds to a study performed on two arid zones in the central zone of Colombia, at Tatacoa and Candelaria deserts, where a microbiological identification of radiotolerant bacteria was made, in order to understand the possible mechanisms of tolerance from life in Mars, stating the hypothesis of these areas as an analogue [11]. Other studies include the development of laboratory equipment to simulate the martian atmosphere and surface UV radiation conditions[12], as well as studies on the development of a space suit simulator.

5. **Methodology**

Colombia provides several types of environment suitable for the development of analogue tests. A list of some selected sites for potential analogue missions is presented. It is important to clarify that no specific mission is suggested to these sites, although their characteristics, and similarities with other locations on Earth, which are already used as analogues, is adopted to select them. Further criteria used to choose them are, their morphological characteristics, their environment type and their accessibility. For all the proposed site the type of environment is defined based on the categorization presented in chapter 2. Also, the main characteristics of each place is outlined along with a proposal of the analogue environment they could represent. Further information and a picture of each site is enclosed.

6. **Potential analogue sites in Colombia**

Figure 1 shows a map with the potential sites identified in Colombia. The marks in orange represent deserts, while the blue marks pinpoint lakes and aqueous environments. Furthermore, the points in purple show

where snow top mountains are located in Colombia. Last but not least, the brown point shows a site near a volcano.

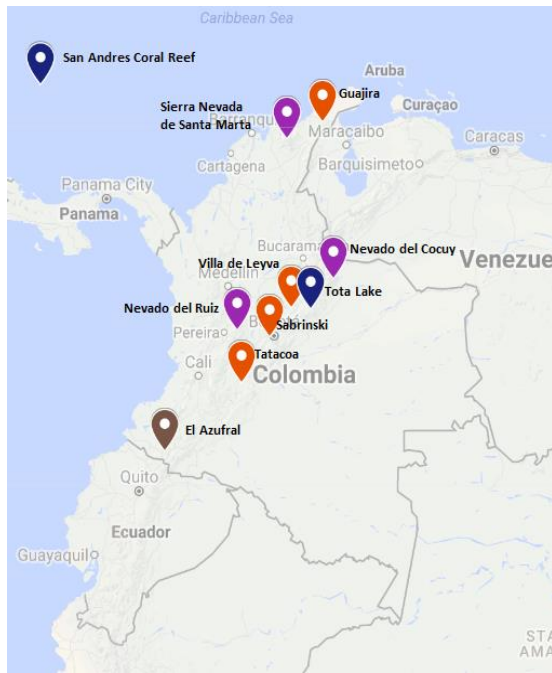


Figure 1: Location of interest sites in Colombia for analogue studies [12].

A List of these sites is presented, as well as some of it's characteristics as analogues.

6.1. Tatacoa Desert

Type: Extreme surface environments - Deserts/arid zone

Characteristics: Heavily eroded and crossed by canyons area. The present landscape offers a variety of landforms as badlands, pinnacles, and isolated erosional remnants forming curious figures produced by surface erosion and the different lithologies affected by the processes [13]. Little light pollution and noise.

Analog of: Planetary surfaces, Regolith zones

Information: Located in the Huila department, comprises an area of 330 km², and annual temperatures between 18 and 43 °C. The desert is also well known for its paleontological research area [14].



Figure 2: Landscape at Tatacoa desert [15]

6.2. Candelaria Desert - Villa de Leyva

Type: Extreme surface environments - Deserts and Caves

Characteristics: Highland dry formations. Multiple natural caves offer moist extreme environments. Little light pollution and noise

Analog of: Planetary surfaces, regolith zones, habitat setup

Information: Located in the Boyaca department, this desert comprises an area of around 100 km², and annual mean temperature around 18 C. As multiple fossils have been found near the village of Villa de Leyva, including one of the worldwide two discovered near-complete skeleton of a Kronosaurus, a Cretaceous-period relative of the crocodile [16].

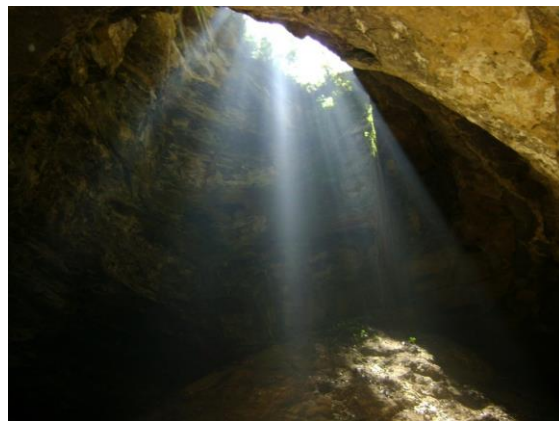


Figure 3: Natural cave near Villa de Leyva [17]

6.3. Guajira

Type: Extreme surface environments - Deserts

Characteristics: This desert has erosional and depositional landforms, which are still under the influence of active processes [18].

Analog of: Planetary surfaces, Regolith zones

Information: Located in the Guajira department, along a part of the western Caribbean coast, this desert has an annual mean temperature of 28 °C. Rainfall occurs mainly during the months of October and November and it is often less than 400 millimeters annually [19].



Figure 4: Sand dunes at Guajira desert [20].

6.4. Sabrinsky

Type: Extreme surface environments - Deserts

Characteristics: Heavily eroded and crossed by canyons area. The reddish and orange sand of this desert, rich in iron and aluminum, is unique in Colombia [21].

Analog of: Planetary surfaces, Regolith zones

Information: Located in the Cundinamarca department, close to Bogotá. Its atypical ecosystem occurred as a response of plant and animal organisms given the conditions of the environment. In particular the influence of the dry winds and few rain. Although, there are few trees low vegetation exists mainly in form of shrubs [22].



Figure 5: Landforms of Sabrinsky desert [23]

6.5. El Azufral

Type: Extreme surface environments - Vulcano or Thermal

Characteristics: As a recent stratovolcano with a complete magmatic evolution, its substrate consists of thick Tertiary pyroclastic deposits with thermal manifestations such as phreatic eruption craters and hot springs. The arid area has remarkable morphological features which is a caldera structure with a diameter of about 3 km, in which there is a green acid lake. The green colour is given due to the sulphur in it [24].

Analog of: Planetary surfaces, Regolith zones

Information: Located in the Nariño department. The volcano reaches a high up to 4700m over the sea level. Temperatures are around 8°C.



Figure 6: Green lake at El Azufral volcano's caldera [25].

6.6. Nevado del Ruiz

Type: Extreme surface environments - Cold desert

Characteristics: Heavily eroded and crossed by canyons area. Surrounding area mostly consists of pyroclastic flows.

Analog of: Planetary surfaces, Regolith zones. Outposts and hardware test.

Information: Located in the Huila department, has a maximum height of 5,321m with its summit covered with glaciers. It still active, with activity as recent as 2016. The zone is part of a national park. The presence of slopes between 10° to 30° makes it suitable for different kind of hardware testing.



Figure 7: Nevado del Ruiz, as seen from the International Space Station.[26]

6.7. Sierra nevada de Sta. Marta

Type: Extreme surface environments - Cold desert

Characteristics: High slopes, arid zones, canyons.

Analog of: Planetary surfaces, Regolith zones

Information: Located in the Magdalena department, is the tallest mountain in Colombia with its highest peak reaching 5.700m. The zone is declared an Indian Reserve, for the presence of several ethnic groups living in the area.



Figure 8: Sierra Nevada de Santa Marta. [27]

6.8. Sierra Nevada del Cocuy

Type: Extreme surface environments - Cold desert

Characteristics: Arid regolith zone, with glaciers. High altitude.

Analog of: Planetary surfaces, Regolith zones, outpost and hardware test.

Information: Located in the Boyacá department, the reserve comprises an area of 3062km², containing several peaks above 5000m, has a maximum height of around 5300m. Accessible by road.



Figure 9: Sierra Nevada del Cocuy. [28]

6.9. Simón Bolívar aquatic center

Type: Underwater confined facilities

Characteristics: Diving pool, up to 5m depth, under roof installation, with visibility from side windows.

Analog of: Low gravity environment, water environment for hardware test.

Information: Located in the city of Bogota, public aquatic complex, ran by the district's institute for sports and recreation.



Figure 10: Diving pool of the Simón Bolívar aquatic complex. [29]

6.10. San Andres Coral Reef

Type: Low depth aquatic environment

Characteristics: Low depth coral reef, with

Analog of: Possible Underwater habitat location, Hardware test.

Information: Located in the Caribbean Sea. Is one of the largest coral reefs in the world.



Figure 11: Aerial view of San Andres island coral reef.[30]

6.11. Tota Lake

Type: Low depth aquatic environment

Characteristics: High altitude, fresh water lake.

Analog of: Possible Underwater habitat location, Hardware test.

Information: Located in the Boyacá department, comprises an area of 55.1km², and average water temperatures of 13°. It's the biggest natural lake in Colombia, located in an elevation of 3.015m.



Figure 12: View of White Beach, on the shore of Tota Lake.[31]

6.12. Colombian Air Force planes

Type: Parabolic Flights

Characteristics: With a concluded peace process between left wing guerillas and the government, much of the military infrastructure available in Colombia will be subjected to operation time reduction, among them,

the Colombian Air Force planes, some of which are suitable for parabolic flight.

Analog of: Low gravity environment

Information: The fleet of the air force includes C-130 Hercules, KC-767, Casa 295, among others.



Figure 13: KC-767 Plane from Colombian Air Force. [32]

7. Conclusions and outlook

The Colombian geography offers multiple possibilities for analogue mission sites as well as diverse existing infrastructure, such as diving pools or air force planes, which could be used for these kinds of mission. Possible analogue sites are identified and sorted based on a general categorization of existing worldwide analogue sites. Each of the identified site is characterised with basic information and a picture, to show the variety of the geography present in Colombia. The identification of this analogues is a first step for the country towards taking the opportunity to expand research in space exploration matters. Future steps shall include a deepened study on each of the sites as well as a trade-off of them, to selected the best suited places for analogue missions in Colombia.

For other studies, facilities such as hospitals, military bases, and medical research institutes can be considered as well, for Bedrest, and confinement studies; this considering that there exist several medicine faculties in the country, with one offering a graduate program on aerospace medicine.

It is also important to note that by performing analogue missions on the described sites, it is also possible to learn more from them, and to generate interest from the general public in their preservation.

Finally, having these sites provides the South American region with a start point to the development of joint efforts towards space research, with resources readily available.

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